

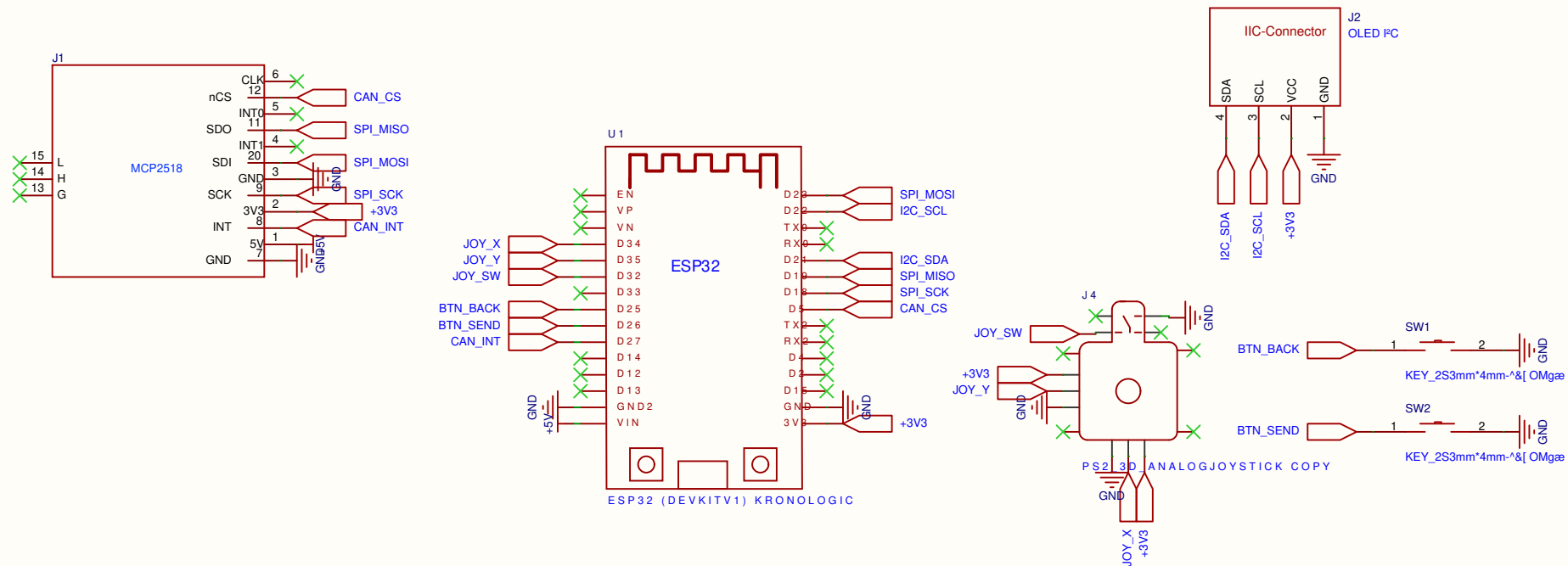
WIRING SCHEMATIC & NET LABELS

Standalone CAN / CAN FD controller reference

ESP32 DevKit V1 (30-pin) • MCP2518FD • I²C OLED • PS2 joystick • two control buttons

DOCUMENT CONTENTS

- 1 Electrical schematic
- 2 ESP32 pin-to-net mapping
- 3 MCP2518FD module connections
- 4 OLED, joystick and button wiring



Schematic	Schematic1			Create at	2026-09-16
				Update at	2026-09-16
Board	Board1			Page	P1
Drawn		esp32 FD			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

U1 • ESP32 DevKit V1 / 30-pin



Pin / function	NET LABEL
VIN / 5V	+5V
3V3	+3V3
GND, GND2	GND
D18 / GPIO18	SPI_SCK
D23 / GPIO23	SPI_MOSI
D19 / GPIO19	SPI_MISO
D5 / GPIO5	CAN_CS
D27 / GPIO27	CAN_INT
D21 / GPIO21	I2C_SDA
D22 / GPIO22	I2C_SCL
D34 / GPIO34	JOY_X
D35 / GPIO35	JOY_Y
D32 / GPIO32	JOY_SW
D25 / GPIO25	BTN_BACK
D26 / GPIO26	BTN_SEND

Dxx and GPIOxx identify the same ESP32 pin. Copy every net label exactly as shown, without spaces. All other GPIO pins are unused in this schematic.

This is a wiring reference, not a PCB footprint drawing. Before PCB fabrication, verify all 30 pins, the spacing between the two header rows and the pad numbering of the selected footprint.

ESP32 photo: [ICShop](#)

J1 • MCP2518FD SPI / CAN FD



Pin / function	NET LABEL
nCS	CAN_CS
SCK	SPI_SCK
SDI	SPI_MOSI
SDO	SPI_MISO
INT	CAN_INT
5V	+5V
3V3	+3V3
GND (both pins)	GND
CLK	NC
INT0	NC
INT1	NC
L / H / G terminal	NC*

* L / H / G are used through the module's built-in green terminal block. No carrier-board traces are routed to these pins. NC means no external connection in this schematic; it does not mean the module has no internal connection.

Do not add a separate CAN terminal block or an additional 120-ohm resistor here. The SPI connector GND pins remain connected to the common ground.

Important: do not transfer pin numbers from an earlier pinout into the footprint. Your symbol uses different numbering. Match every pad to the printed label on the actual module. Keep the power and jumper configuration exactly as in the verified working assembly.

Photo supplied by the user (Jessinie)

The product page could not be independently read; the supplied photo and schematic are the basis of this table.

OLED • PS2 JOYSTICK • BUTTONS

J2 • I²C OLED

OLED photo: Robu

Pin / function	NET LABEL
GND	GND
VCC	+3V3
SCL	I2C_SCL
SDA	I2C_SDA

VCC and GND order differs between OLED modules. Follow the markings printed on your board.

J4 • PS2 analog joystick



PS2 photo: OnBuy (example)

Pin / function	NET LABEL
X-axis wiper	J0Y_X
Y-axis wiper	J0Y_Y
X/Y outer terminals	+3V3 / GND
Switch: one contact	J0Y_SW
Switch: other contact	GND

This is not a KY-023 module: the schematic uses the bare PS2 joystick mechanism. Each potentiometer has three pins; the center pin is the signal. Verify the orientation of the outer pins against the schematic. Do not assign mounting tabs to GND without testing.

SW1 + SW2 • Two buttons



Pin / function	NET LABEL
SW1: contact 1	BTN_BACK
SW1: contact 2	GND
SW2: contact 1	BTN_SEND
SW2: contact 2	GND

The photo is an example of a two-contact switch, not confirmation of the KEY_2S3mm*4mm package. Verify the footprint against your actual buttons. Pressing the button shorts the signal to GND.

Button photo: Componentes Electrónicos CR